M KIRTHIK ADITYA

Aspiring Security Engineer

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PROFESSIONAL SUMMARY

Cybersecurity student interested in penetration testing, network security and cloud security. I learn by doing: CTFs on TryHackMe and PwnTillDawn, a Proxmox homelab, and writing security tools in Go, C++ and Python. I work mostly on Linux with Docker and AWS, and I enjoy contributing to open source.

PROJECTS

BIT CyberSpace — College Cybersecurity Lab Platform (team project; my role: infrastructure & deployment)

Technologies: Linux, Docker, Nginx, Cloudflare Tunnel, MySQL, Go | cyberspace.skysec.org

- Ran the production server, with the API running under systemd as an unprivileged service user.
- Published cyberspace.skysec.org through Cloudflare Tunnel, keeping the server behind the college NAT with no inbound ports; lab PCs reach it directly over the LAN with Let's Encrypt TLS.
- Configured nginx to allow the student assessment API only on the LAN listener and only from the lab subnet, checked by listener and source address rather than spoofable headers.
- Hosted the Docker lab runtime: one container per student on a macvlan network with a fixed IP, with cleanup that only touches platform-labeled containers.

PurgeMark — NIST SP 800-88 Drive Sanitization, Bootable Linux ISO

Technologies: C++, Linux, NVMe, OpenSSL, Debian live-build | purgemark.skysec.org

- Built a bootable Debian ISO in C++ that erases drives with their own NVMe and ATA commands through the kernel's NVMe admin and SG_IO interfaces, falling back to an AES-256-CTR overwrite.
- Verifies every erase by reading ~1,000 random regions back with O_DIRECT, and plants canaries before firmware erases to catch drives that fake success.
- Issues Ed25519-signed certificates over RFC 8785 canonical JSON, with a CLI verifier and a browser verifier that never uploads the file.
- Refuses mounted/swap/LVM/RAID/LUKS disks; tested in Proxmox VMs with a destructive integration test that picks disks by serial number.

IndexCase — Malware Propagation Tracer for Active Directory (in progress)

Technologies: Python, Polars, Parquet, Windows Event Logs

- Reads EVTX and JSONL logs from many hosts and normalizes Security, System and Sysmon events into one Parquet schema.
- Resolves IPs to hosts over time, preferring Sysmon telemetry over client-supplied workstation names, which Impacket can fake.
- Next: linking remote execution (PsExec-style SMB + service, WMI, scheduled tasks) to its causing logon and ranking the likely patient zero.

TECHNICAL SKILLS

Programming	Go, C++, Python, Bash Scripting
Cloud	AWS, IAM & Access Control
Containers & DevOps	Docker, Linux, Git/GitHub
Security	Wireshark, Burp Suite, Nmap, Metasploit, Vulnerability Assessment, Penetration Testing
Networking	TCP/IP, DNS, HTTP, OSPF, Packet Analysis

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

Bannari Amman Institute of Technology | 2024 – 2028

CERTIFICATIONS & AREAS OF INTEREST

- Cisco – Introduction to Cybersecurity
- Pwn Till Dawn — online.pwntilldawn.com/Achievements/9259 — #13 in India, #466 on the battlefield, 1,315 points, 8 machines

Interests: Cloud Security | DevSecOps | Network Security | Cybersecurity Engineering

CTF & SECURITY COMMUNITY ACTIVITY

- Active on tryhackme.com/p/SPRYZZEN — solving rooms and CTF-style challenges across web, network, and cloud security tracks.
- Publish cybersecurity and cloud security write-ups on medium.com/@mkirthikaditya.al24, translating hands-on lab and project work into technical blog posts.
- Build security tools on github.com/spryzz3n, from drive sanitization in C++ to incident-response tooling in Python. First open-source contribution: a merged Zeek fix that stops its raw input reader looping forever on an empty record separator.